CV2X Indoor Localization with WiFi Fingerprints

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NYCU CV2X Lab 3

Four required sections

Problem · Method · Dataset & splits · Results

- 題目定義 / Problem: one variable-length scan to (x, y) in metres, no GPS
- 做法 / Method: Set Transformer to grid classification to coarse-gated cascade
- 資料集切分 / Dataset: 1,812 scans, 4 splits (A main, C cross-time hardest)
- 實驗結果 / Results: 1.57 m to 0.75 m, validated on a held-out test set + live hardware

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題目定義 / Problem Definition

What exactly are we predicting, and why is it hard?

The task: turn one variable-length WiFi scan into a point on the map

Input = a set of (BSSID, RSSI); output = (x, y) in metres

- Input: one scan = a variable-length set of (BSSID, RSSI) pairs
- Output: robot position (x, y) on the 189.5 m^2 floor plan
- Four difficulties: variable set, ambiguous RSSI, sparse coverage, 0.3 m label floor

Anatomy of one scan: a ragged bag of (BSSID, RSSI)

Same place, two scans, different lengths — order means nothing, presence means everything

- One scan = an unordered bag like {(ap_3f, -52), (ap_1a, -67), (ap_b2, -78), ...}
- Length varies scan-to-scan (~12 to ~40 APs); same spot need not list the same APs
- Force it into an 80-D vector and ~2/3 of it is -100 padding (only 27.7 of 80 slots are real APs)
- Order carries no meaning; only which APs are present and how strong they are

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資料集切分 / Dataset & Splits

1,812 real scans, a coverage hole, and four generalisation stress-tests

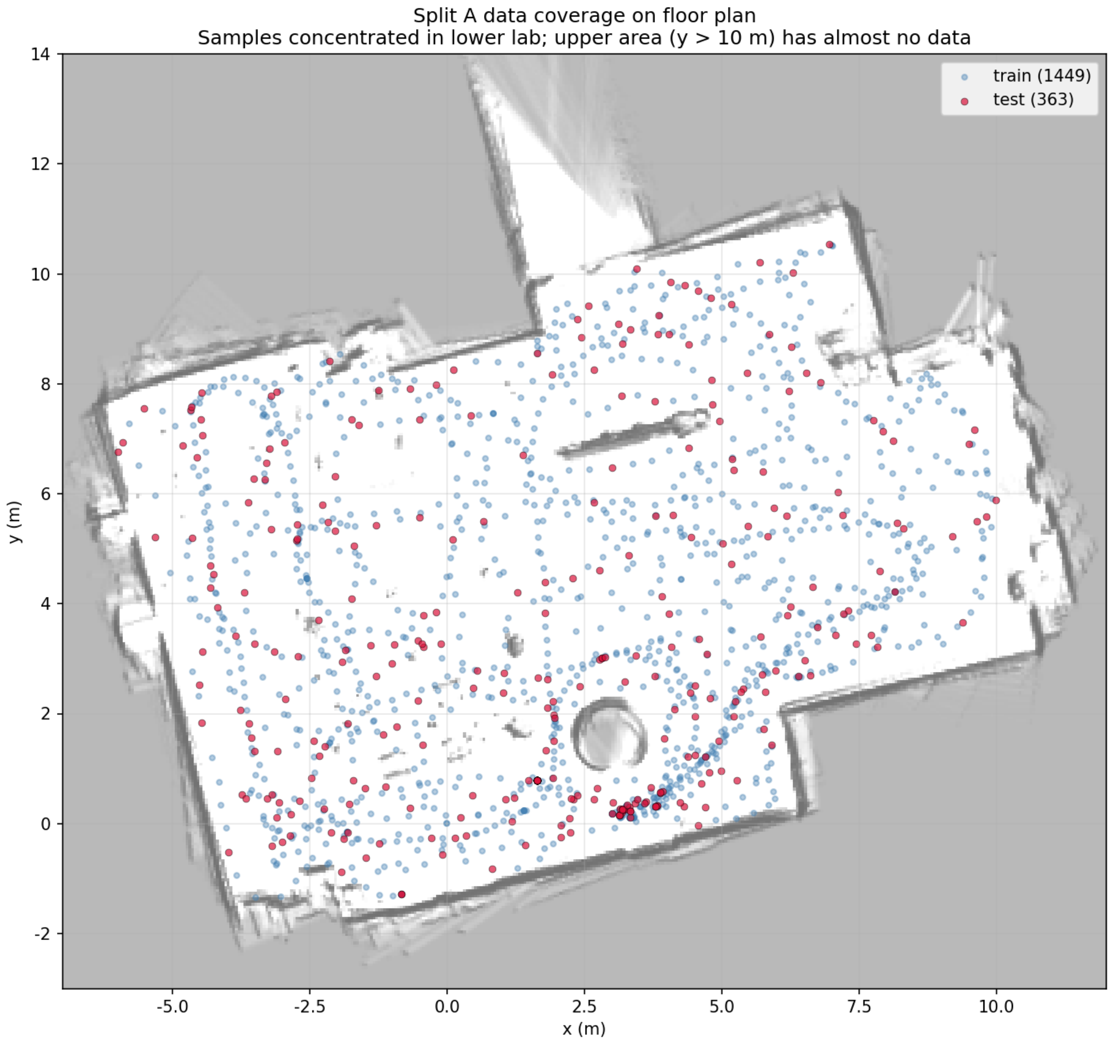
Every scan is a real robot run: ESP32 on a TurtleBot3 doing SLAM

Hardware-collected data, not simulation — and that dictates every dataset quirk

- ESP32-S3 sniffer rides a TurtleBot3 running Cartographer SLAM + AMCL
- Each scan ~3 s; ground truth (x, y) is AMCL's pose (~0.3 m noise)
- Coverage = wherever we drove — path-shaped, not a uniform grid
- Two sessions (morning + evening) to expose later cross-time drift

1,812 fingerprints in a 189.5 m² lab — but the upper region is a coverage hole

- 1,812 scans = morning 912 + evening 900; train/test split 1449/363
- 115 BSSIDs seen, 80 kept (>=10 hits); the rest are transient noise
- Ground truth from ROS AMCL, ~0.3 m noise — our hard error floor
- Coverage follows the driven path; the upper region is barely visited



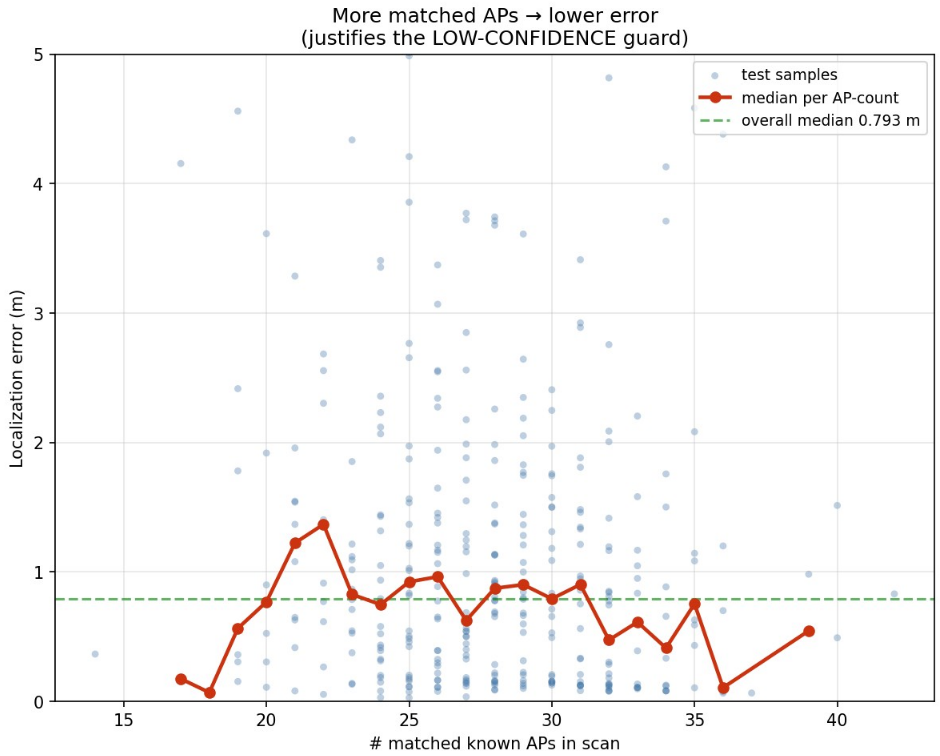
Four splits, each stress-testing a different generalisation axis

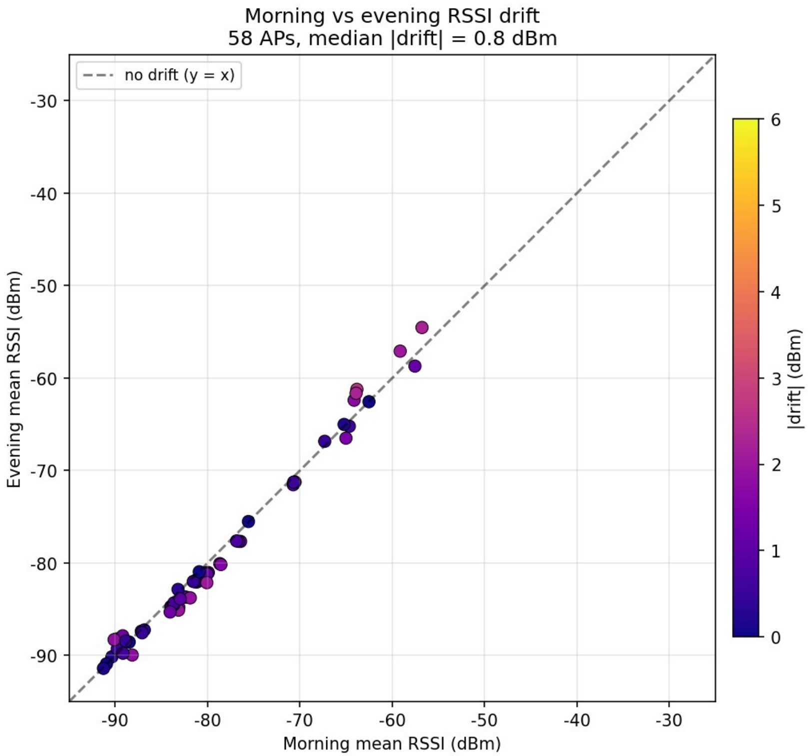
A_random is the main number; C (cross-time) is the hardest

- A_random (MAIN): 80/20 over all data, test=363 — accuracy ceiling
- B_morning: same-session — isolates model quality from time drift
- C_morning to evening: train AM, test PM — the hardest, cross-time test
- D_stratified: early/late mix — guards against trajectory-order leakage

Coverage, not noise, is the real bottleneck

- Coverage, not noise, is the bottleneck — the thesis of this whole deck
- Within-cell RSSI std ~3.5 dBm; morning-to-evening drift only ~0.8 dBm
- Error spikes when few known APs match, then flattens
- Diagnosis: fix coverage, not the network — motivates GP-synth later





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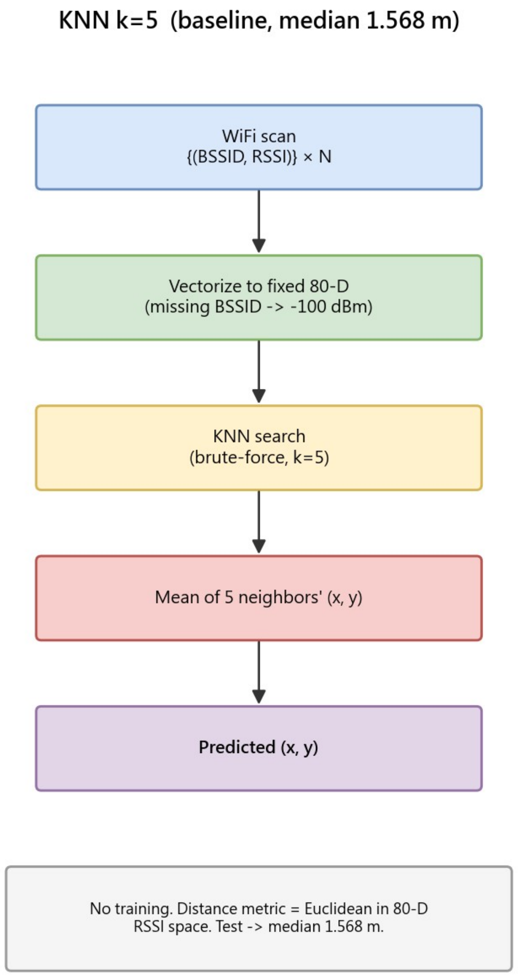
做法 / Method

The climb: each method change fixes one named problem

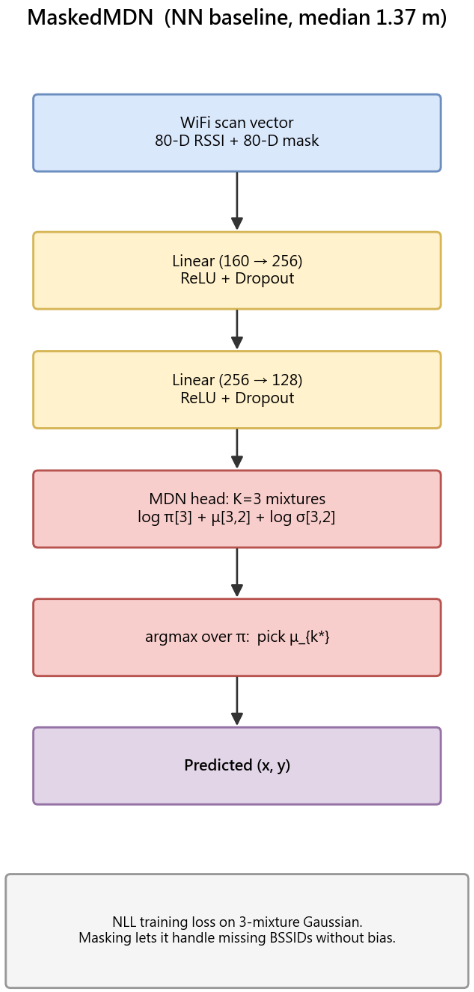
Classic KNN fingerprinting sets the bar to beat at 1.57 m

- Scan to 80-D RSSI vector; unheard AP padded to -100 dBm (not heard)
- k=5 (a-priori, not tuned on test) nearest neighbours, distance-weighted pose
- Zero training, pure lookup: the reference point every net must beat
- Median error 1.57 m on Split A test



Plain neural nets clear KNN but plateau near 1.3 m — two lessons

- MLP 1.30 m and MDN 1.26 m: both about -17% vs KNN 1.57 m, yet barely apart from each other
- A mixture head (MDN) only inches past the MLP; masking the -100 padding (1.37 m) made it worse, not better
- Lesson 1 - representation: a fixed 80-D, -100-padded vector wastes most of the model capacity
- Lesson 2 - output: a point/mixture readout still averages two plausible places - it needs a grid



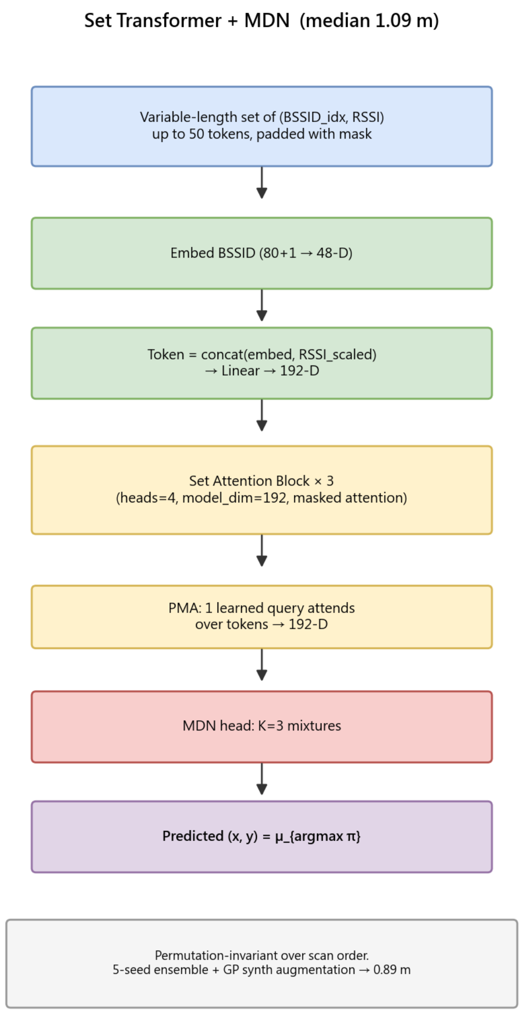
Why predict a distribution, not a single point

One scan can map to two plausible places — a point estimate splits the difference

- Symmetric layouts make one RSSI fingerprint match two real positions
- L2 regression's optimum is the mean — it lands in the empty middle
- MDN expresses multiple peaks but its readout still averages them
- Cleanest fix: classify over map cells, where peaks stay separate

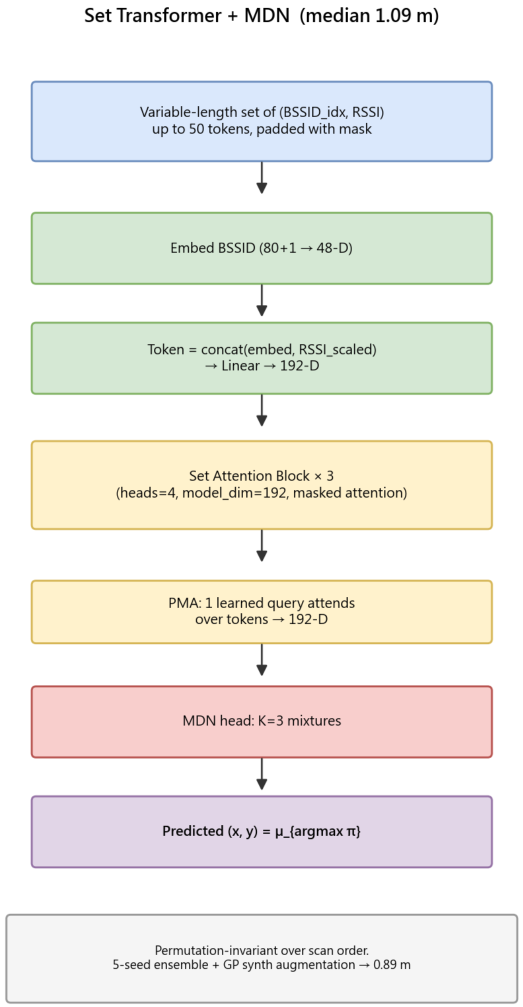
Treating the scan as a SET, not a vector, unlocks 1.09 m

- Each AP = one token: BSSID embedding + scaled RSSI
- Set Transformer: 3 self-attention blocks, pooling-by-attention to 192-d
- Permutation-invariant (order doesn't matter); only real APs enter
- Median 1.09 m, -16% vs the MLP



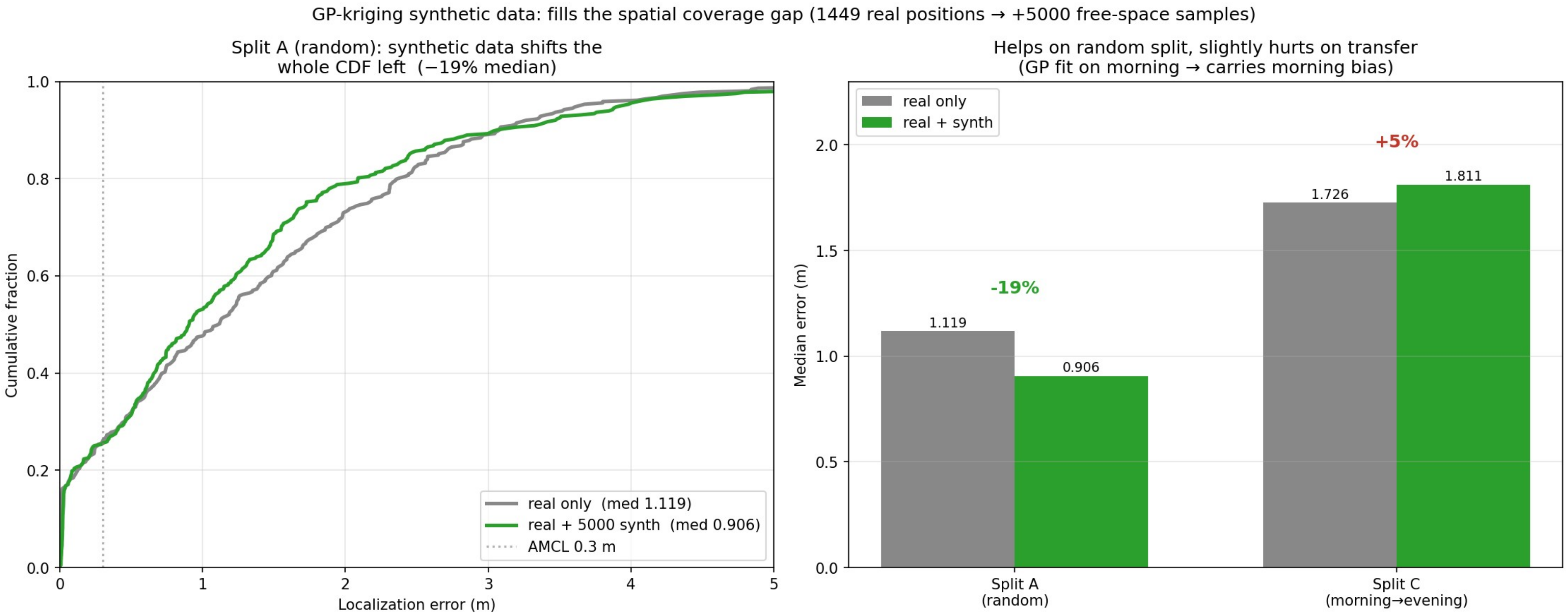
How a Set Transformer reads a variable-length set

- Each (BSSID, RSSI) becomes a token: learned ID embedding + scaled RSSI
- Self-attention = APs compare to each other, building joint context
- Permutation-invariant: no positional encoding, order carries no meaning
- PMA: a learned query attends over all tokens to a single 192-d vector



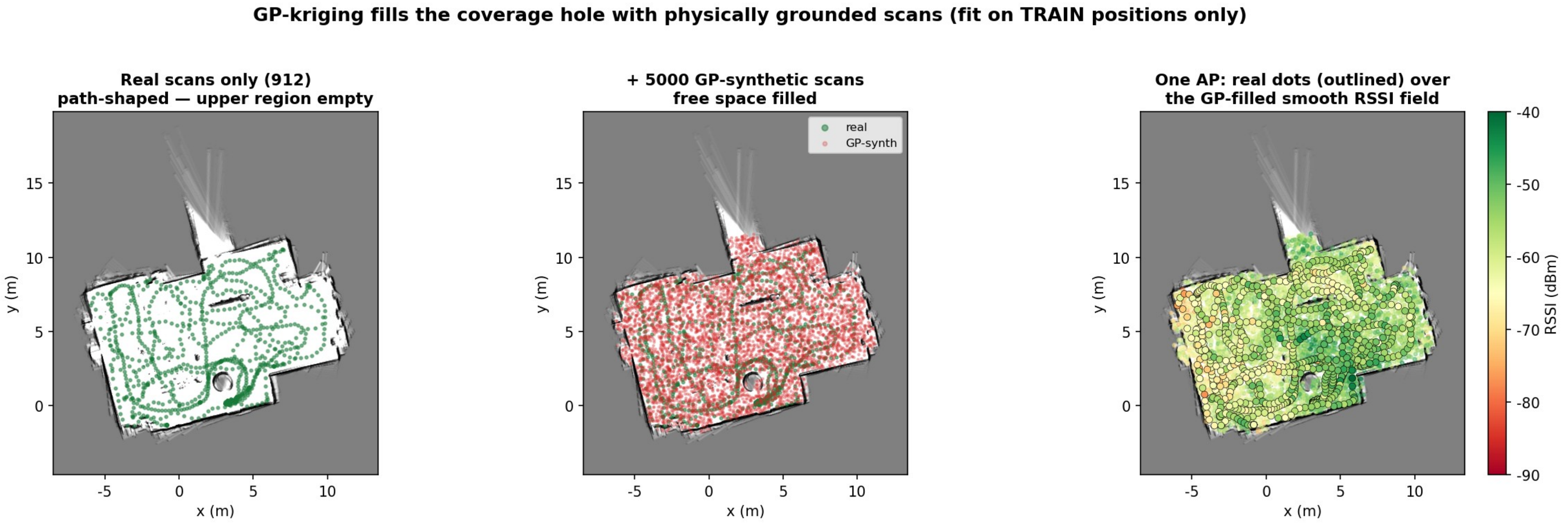
GP-kriging synthetic data: -19% to 0.91 m by filling the coverage hole

- Per-BSSID Gaussian Process learns a smooth position-to-RSSI field
- Sample 5000 synthetic scans from the empty upper region
- Same-model ablation, Split A: real-only 1.12 m -> +synth 0.91 m (-19%)
- Cross-time (Split C) is +5% worse - the GP carries the morning bias



How GP-kriging fabricates believable coverage

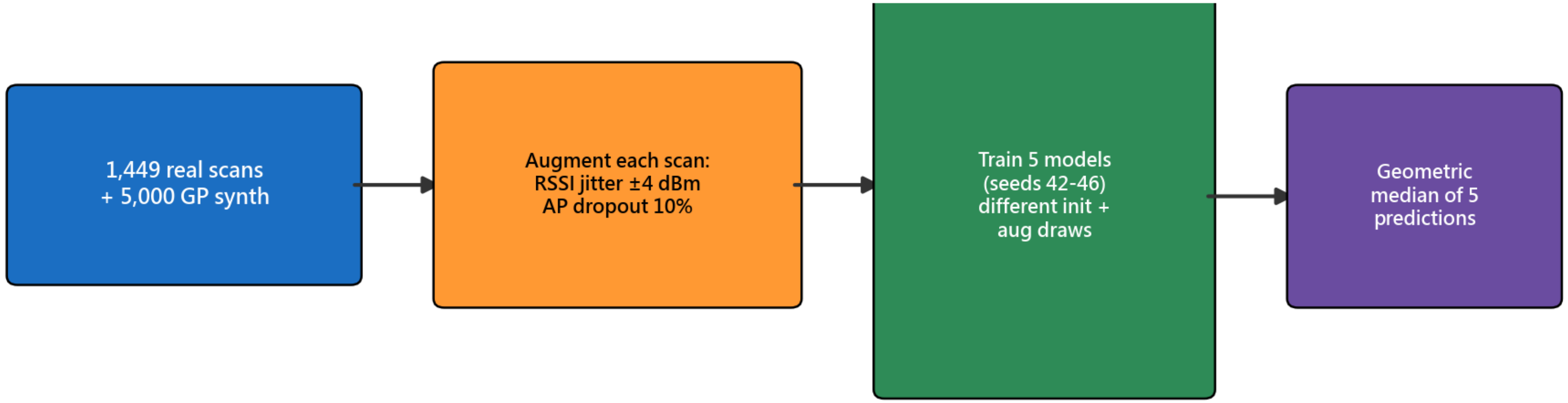
- One 2-D GP per AP maps position to a smooth RSSI field with uncertainty
- A KNN detection model decides whether each AP is even visible here
- Both models are fit on TRAINING positions only — never on test poses
- Sanity check: synth mean 27.1 APs/scan vs real 27.7 — well-matched



The training recipe: augment, 5 seeds, geometric-median ensemble

- Train on real + 5000 synthetic scans together, not in stages
- Augment: RSSI jitter +-4 dBm and 10% random AP dropout per scan
- Train 5 seeds; combine with a geometric-median ensemble (outlier-robust)
- Same recipe across every neural milestone — a fair, fixed comparison

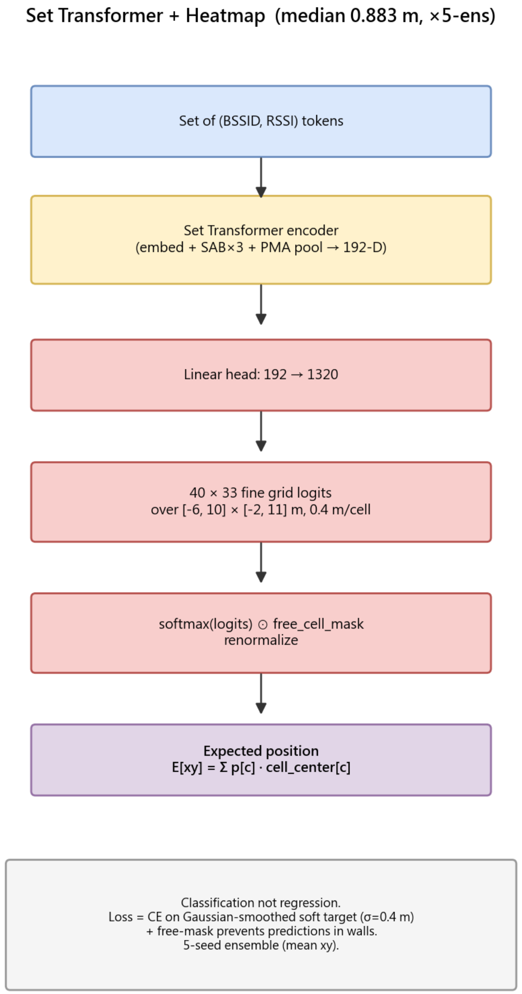
Training recipe: synthetic coverage + noise augmentation + robust ensembling



Why geometric median? It is outlier-robust: one bad seed cannot drag the ensemble, unlike a plain mean. Augmentation injects the sensor noise the model must tolerate.

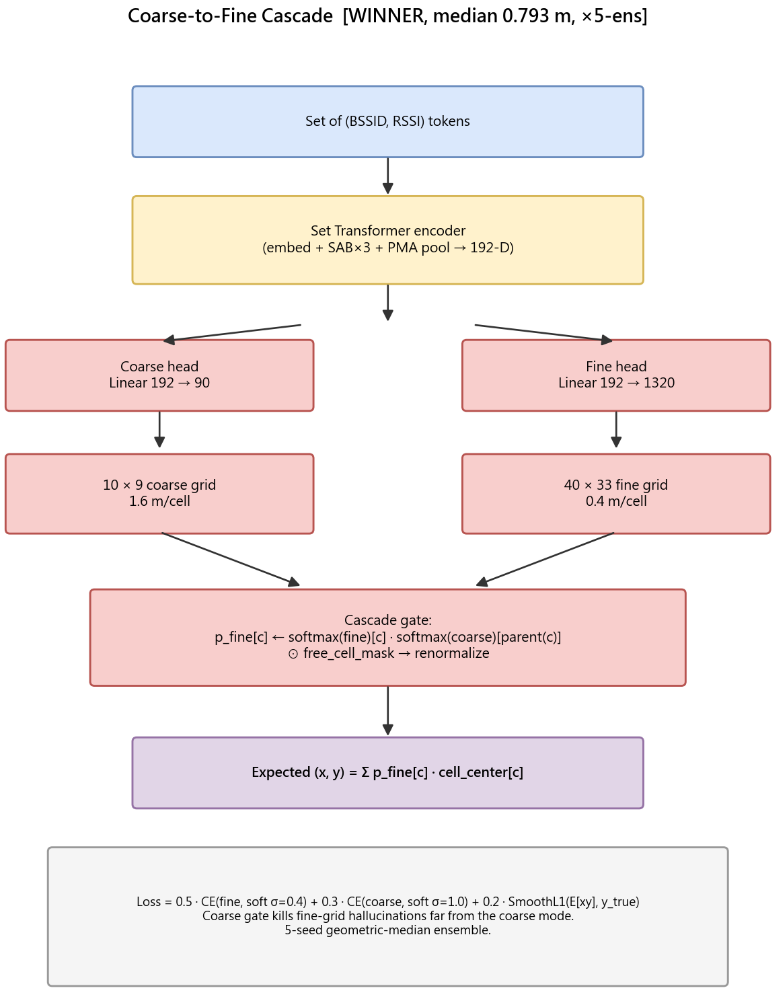
Classifying on a masked grid beats regressing a coordinate: 0.88 m

- Predict P(cell) over a 40x33 grid (~0.4 m cells), not a coordinate
- Free-cell mask zeroes walls; centroid reads off the position
- Classification can express multiple peaks; regression cannot
- Median 0.91 to 0.88 m (5-seed ensemble)



Our champion: a coarse 10x9 gate over the fine 40x33 grid lands 0.79 m

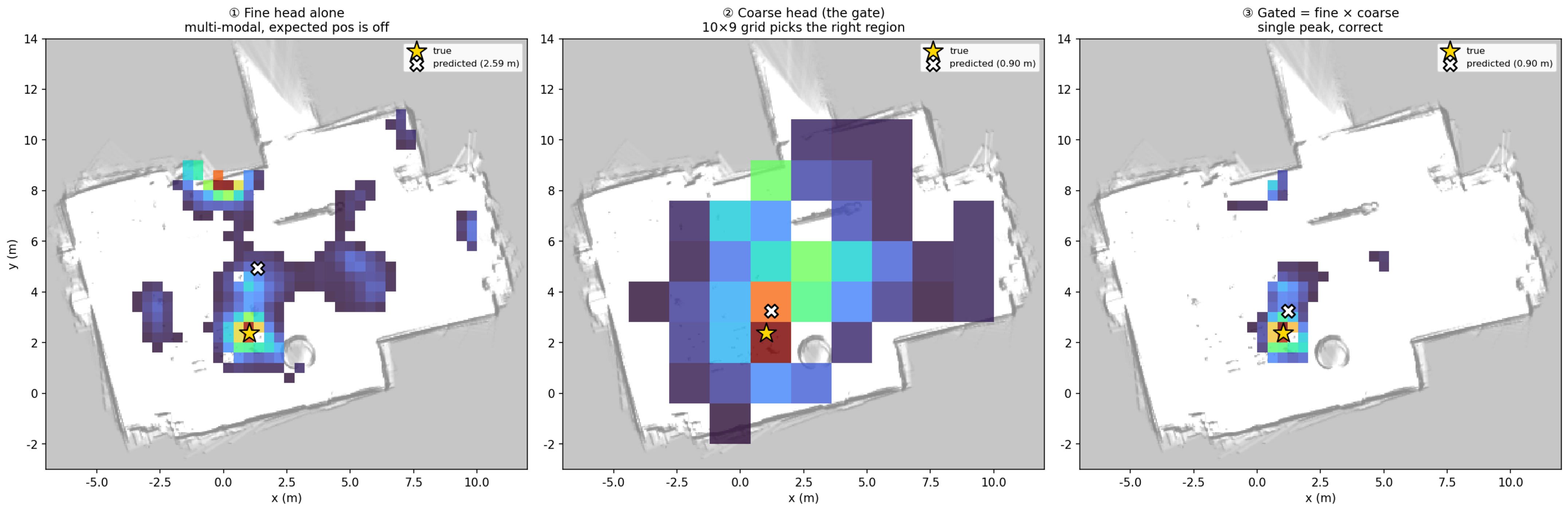
- Shared encoder feeds a coarse 10x9 + a fine 40x33 head
- Gate: P_fine(cell) x P_coarse(parent region) before the centroid
- 0.79 m median (5-seed) — best of every config we tried
- Loss-weight tuning (mse_w 0.55) to 0.752 m — the reported headline



Why it works: the coarse gate kills the fine grid's false second peak

- Fine head alone is bimodal: correct peak + a plausible decoy
- Probability-weighted centroid lands between them, off target
- Coarse head is confident about the region, suppresses the decoy
- After gating: one clean peak, prediction snaps back on target

Why the cascade wins: the coarse gate removes fine-grid ambiguity (this sample: 2.59 m → 0.90 m)



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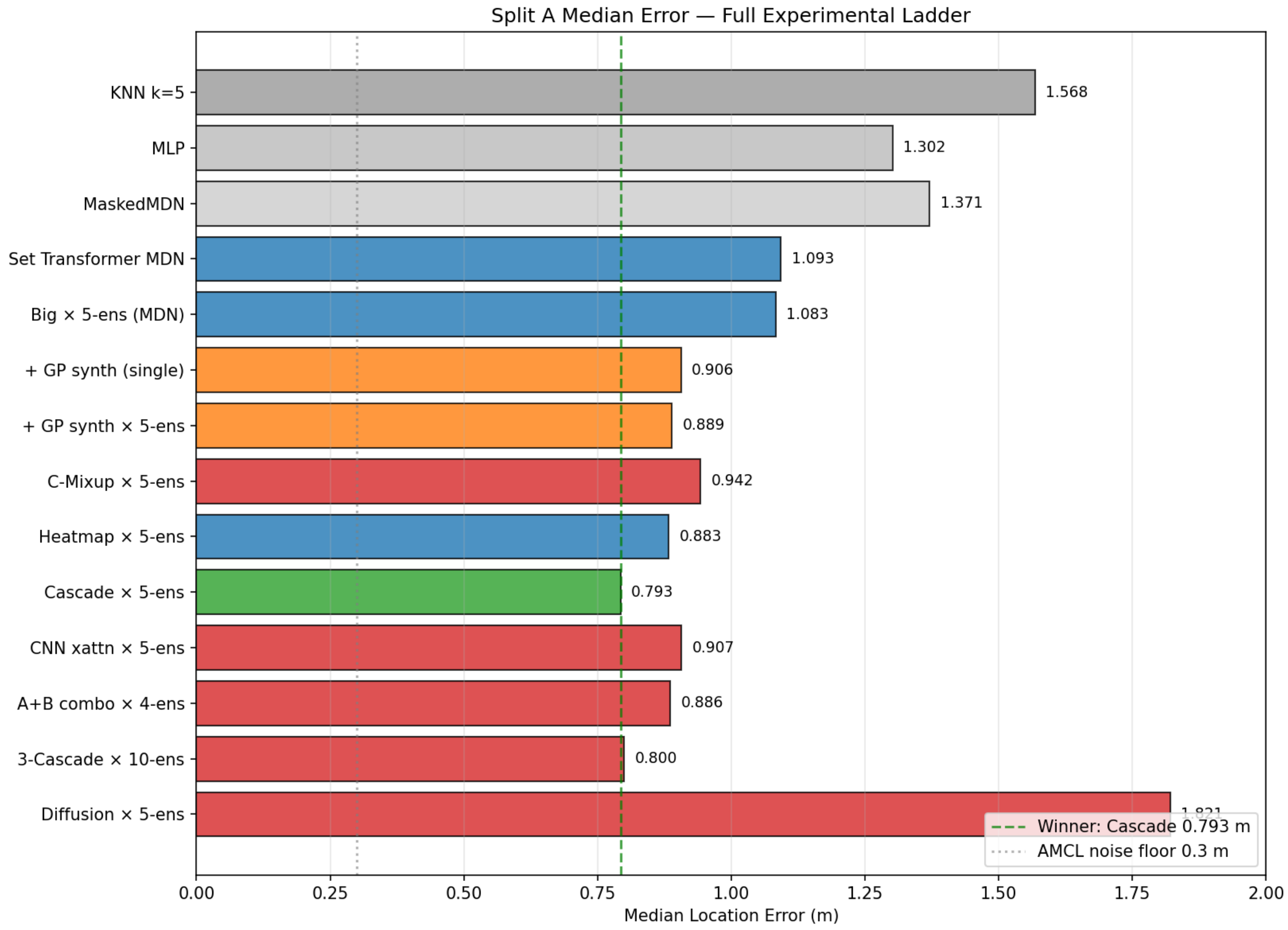
實驗結果 / Experimental Results

The full climb, the validation, the failures, and the live demo

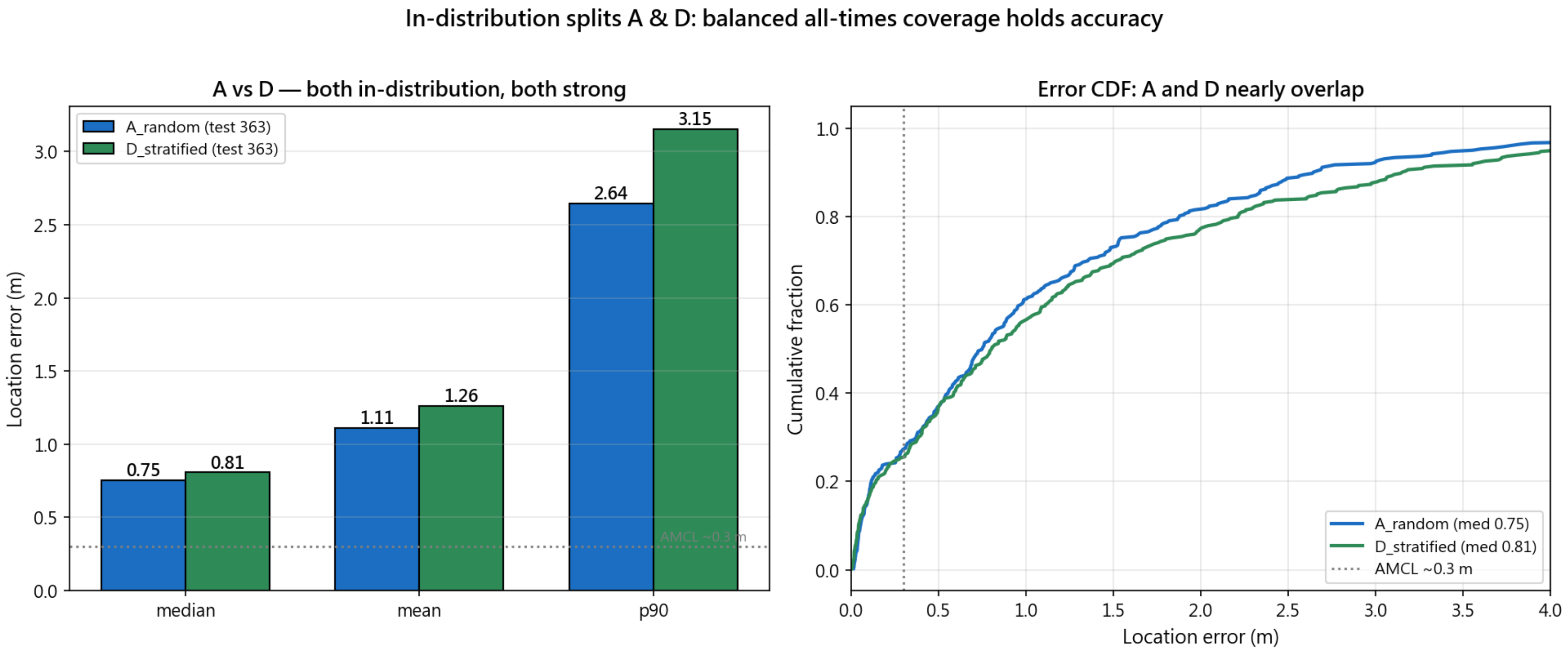
The full climb: every method change bought real metres, 1.57 to 0.75 m

- Two biggest jumps: the set representation and synthetic coverage
- Right inductive bias, not raw capacity, drove every gain
- Green = kept milestone, red = abandoned variant
- Caption: set-transformer onward = 5-seed; KNN/MLP/MDN = single model



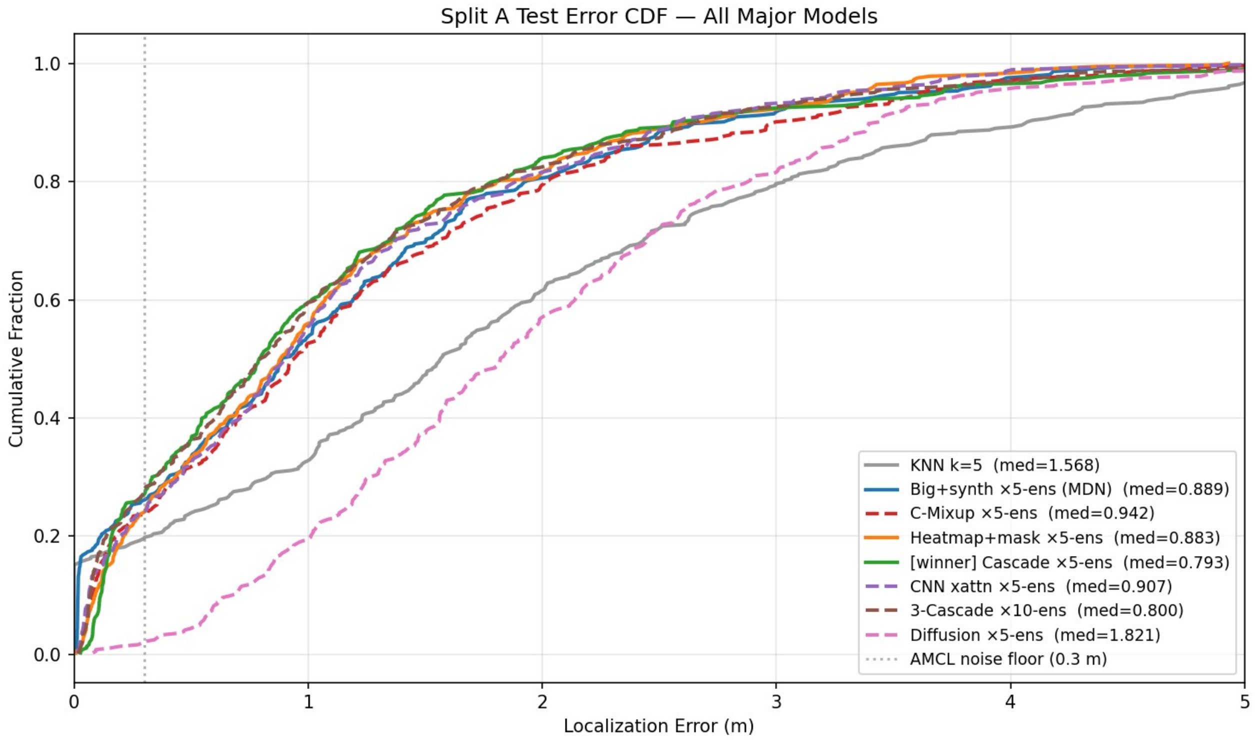
In-distribution splits A & D are both strong (~0.8 m); cross-time C is the open problem

- A_random (MAIN): 0.752 m measured — random 80/20
- D_stratified: 0.807 m measured — both times of day in train & test
- B_morning: ~1.0 m (estimate; morning-only, so less training data than A)
- C_morning to evening: ~2.3 m (estimate) — cross-time, the open problem



The whole distribution improved, not just the median

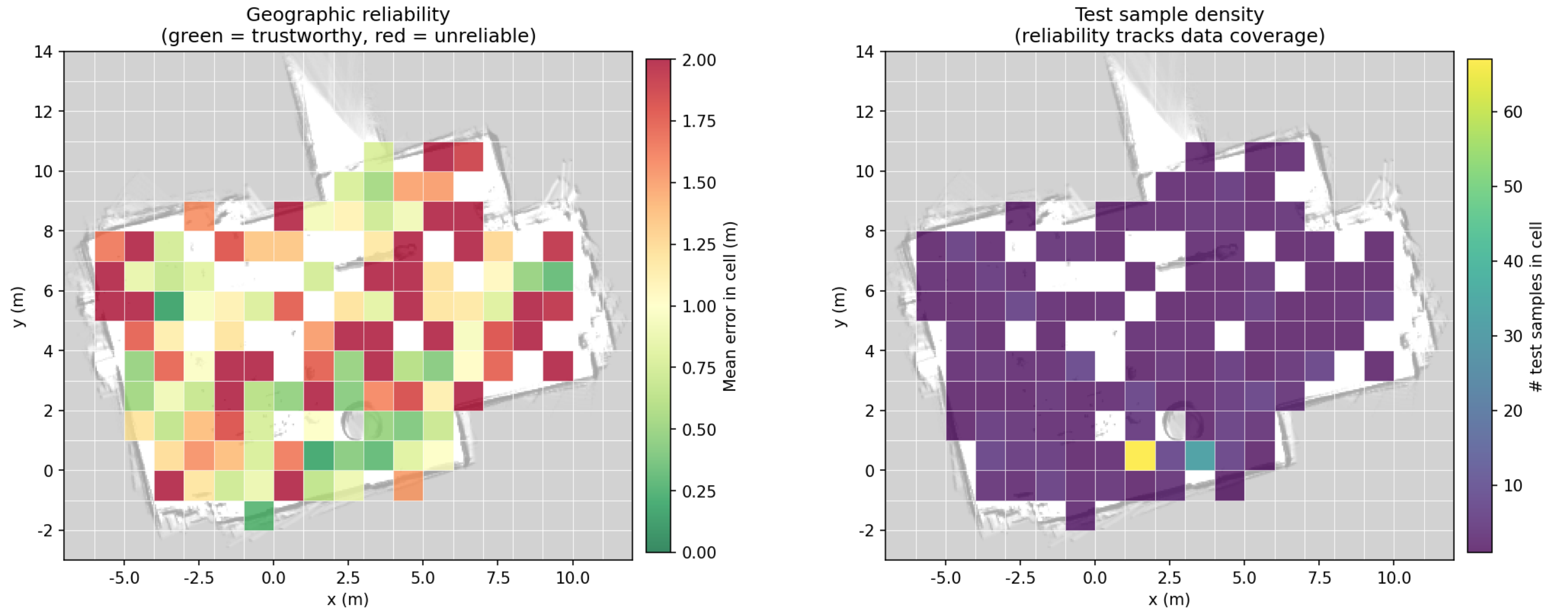
- Cascade CDF sits left of every other model at every percentile
- 27% of predictions within the 0.3 m AMCL floor
- Median 0.79 m, mean 1.12 m, p90 2.56 m
- >2 m tail from sparse regions + symmetric-layout ambiguity



Where to trust it: reliability tracks data density, not network size

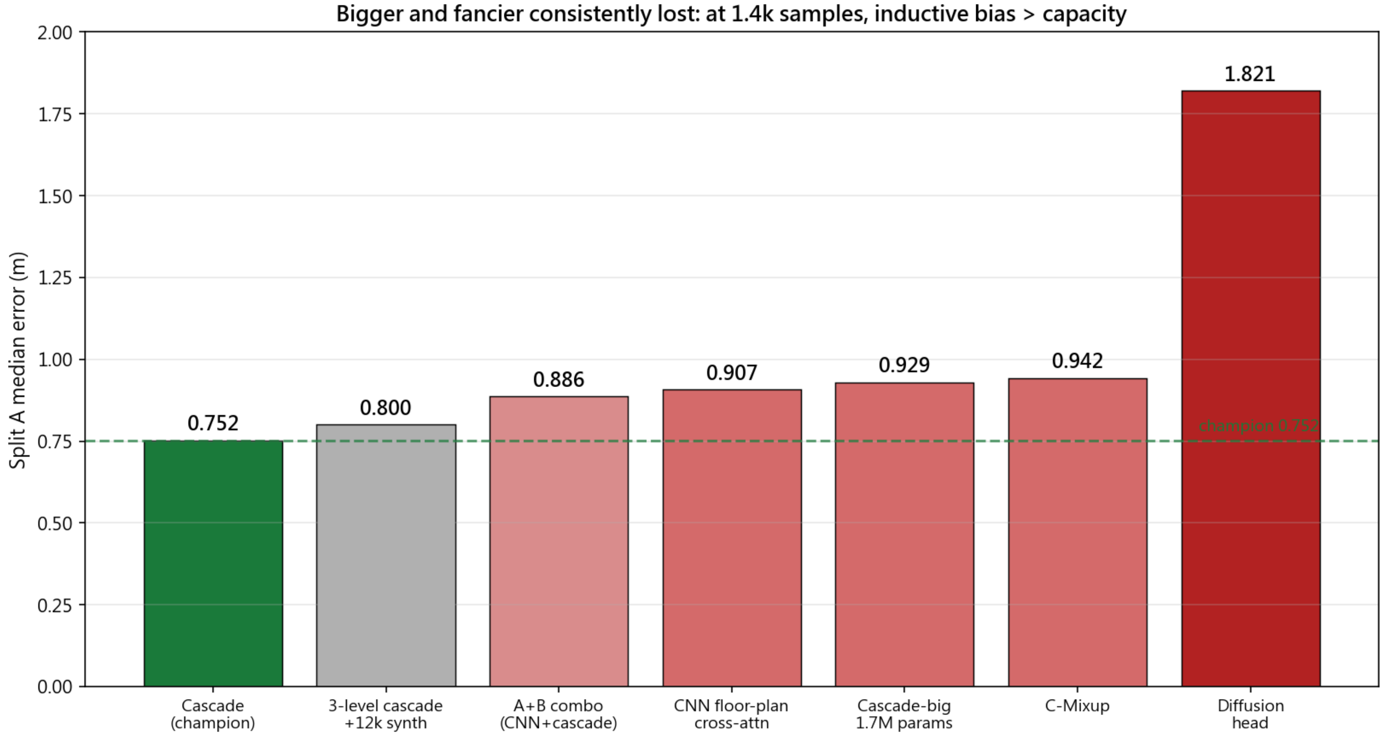
- Per-cell mean error mirrors the sample-density map almost exactly
- Dense, well-travelled regions: well under 0.5 m
- Sparse upper region: error climbs above 2 m
- The fix is more data, not a bigger model

Where can you trust the model? Error is low where data is dense, high in the sparse upper region.



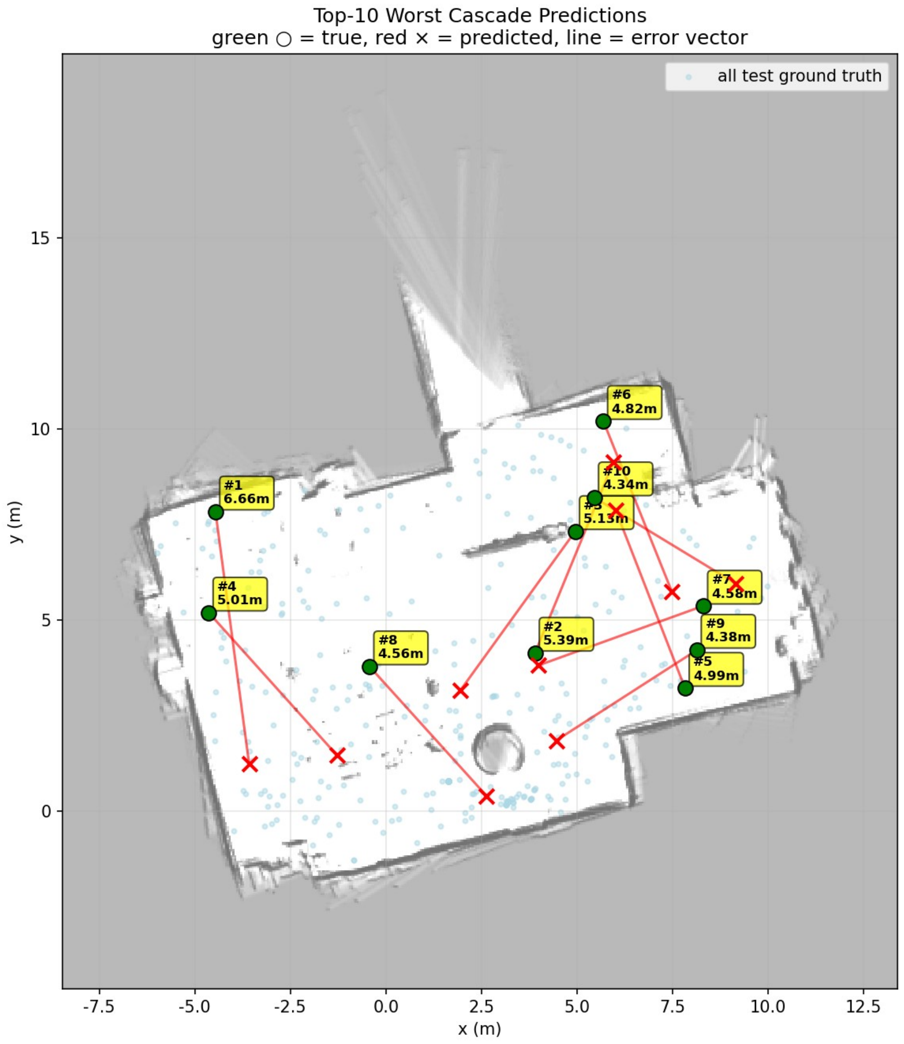
Bigger and fancier consistently lost: at 1.4k samples, bias beats capacity

- Diffusion head 1.82 m: sampling noise on a deterministic task
- Cascade-big 1.7M params 0.93 m: capacity just overfits at this scale
- CNN floor-plan cross-attn 0.91 m: extra machinery, no extra signal
- (also: C-Mixup 0.94, A+B/CNN+cascade combo 0.89, 3-level+12k synth 0.80)



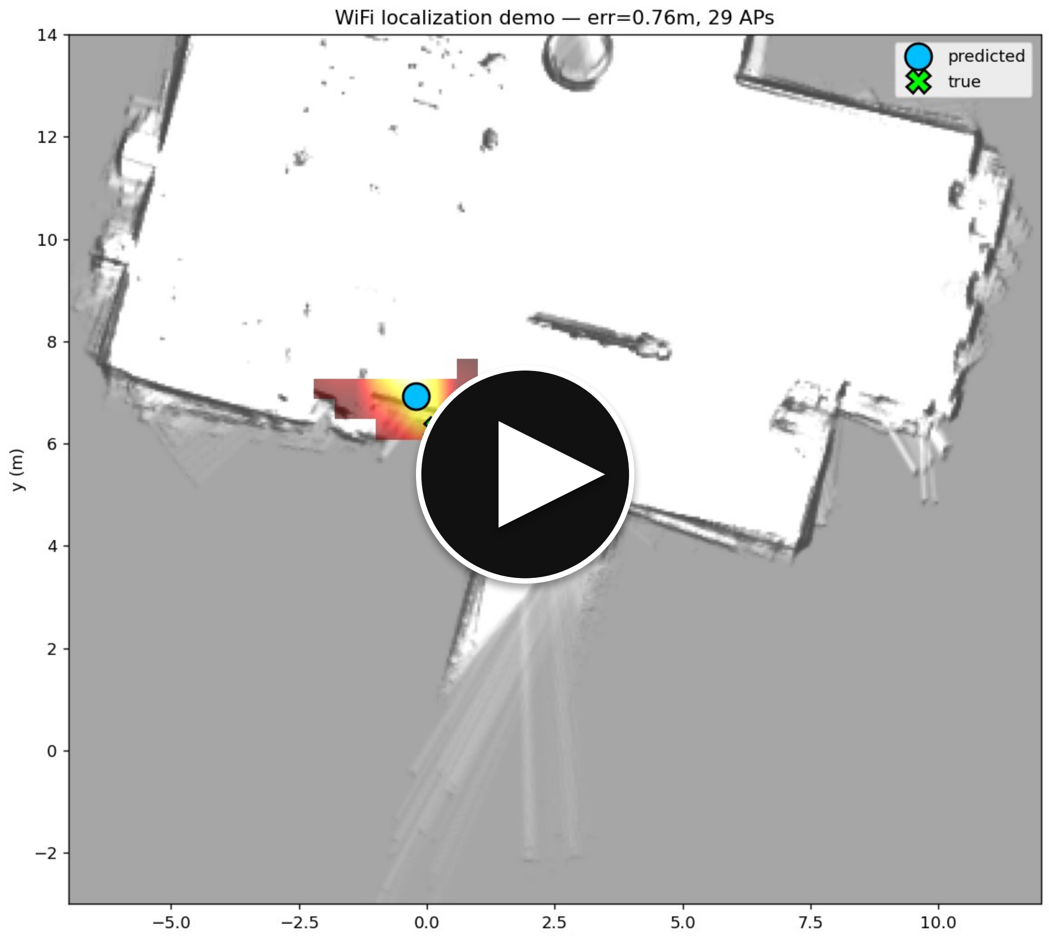
The error splits cleanly: 27% solved at the label floor, 16% unsolvable

- 27% of predictions sit at the 0.3 m AMCL floor (label-limited, not improvable)
- ~16% are >2 m outliers from WiFi symmetric ambiguity (two places, same RSSI)
- Worst-10 cluster in the sparse upper region and mirror-symmetric spots
- No model can separate identical fingerprints — only new signals can



Live in the lab: ESP32 to 9 ms CPU inference to live heatmap

- Real ESP32 streaming live scans; CPU-only inference ~9 ms/scan (no GPU)
- Live position + probability heatmap rendered on the floor plan
- 80-95% of live APs match the trained vocabulary (no drift)
- Precision standing still <0.1 m spread; accuracy vs ground truth 0.4-0.8 m



▶ LIVE DEMO VIDEO — replace this slide image with the recording

Verdict: ~0.75 m is real; sub-0.65 m needs new data, not a deeper net

- Halved KNN error to a 0.752 m median, validated on real hardware
- Won by inductive bias (set input, grid classification, coarse gate), not capacity
- (btw) an early run hit 0.65 m but was tuned on the test set — it does not generalise, so we report 0.752 m
- To break 0.6x: upper-region data, IMU/heading, temporal fusion — not depth